AGL202607071 · D = 1
Small tier: 200,000 meters / 500 feeders · Assigned identifiers end in 1 (practical: feeder 1).
Task 4 incident: South African region unreachable for six hours during billing.

Primary user: a field revenue officer on a phone, limited to assigned feeders and approved visit details. Three decisions govern the platform: monthly field prioritization, outage awareness within 15 minutes, and daily estimated-bill / disconnection review.

Table 1 · Workload inventory

Workload / pattern	Sensitivity	Operated by	Model / reason
Readings ingestion and quality Continuous; bursty arrivals	Pseudonymized household consumption; personal	Data engineer: rules; provider: runtime	PaaS: managed queue and transforms; replay without a bespoke cluster
Lake and SQL reconciliation Nightly batch; monthly query	Linkable consumption; restricted	Data engineer; provider storage/query operations	Managed storage + PaaS query; preserve proven loss analysis
Operational outage alerts Event-time; ≤15-minute target	Meter codes / operational events	App engineer + operations controller; provider runtime	PaaS event processor in Nigeria, independent of foreign lake
Field dashboard and visit API Interactive, business hours	Assigned customer references / evidence	App engineer; platform owner	PaaS-style app deployment; Docker and self-hosted Dokku for teaching
Billing evidence and appeals Daily; interactive review	Personal billing disputes	Billing officer + app engineer	PaaS application/database; preserve human decisions and audit
Loss prediction and help Monthly training; short queries	Feeder aggregates; restricted inputs	Data engineer; revenue lead review	Managed batch PaaS; bounded help first; no autonomous enforcement
Identity re-identification key Low volume; case-authorized	Highest confidentiality; identity linkage	Designated Sunbeam engineer in Abeokuta	Controlled on-premises infrastructure; refuse foreign relocation
Staff identity / support tooling Interactive, routine	Staff accounts / support records	Provider service; Sunbeam account admin	SaaS: reduce undifferentiated operations; MFA and least privilege

Table 2 · Shared responsibility: readings pipeline

Sunbeam keeps	Managed provider operates
Lawful purpose, transfer assessment, DPO oversight, access roles, key separation, retention and deletion	Contracted physical facilities, infrastructure security, host maintenance and managed-service runtime
Ingestion code, schema, unit normalization, deduplication, late-event policy, mapping reconciliation and quality gates	Contracted queue/storage/query availability and scaling mechanisms; service patching
Alert thresholds, incident response, budget limits, backup configuration, restore tests and evidence of service performance	Contracted logging / backup features and fault handling; Sunbeam must enable and verify them

Self-hosted Dokku shifts developer workflow, not underlying host responsibility. No production readings pipeline is deployed by this app. 200,000 × 96 = 19.2 million readings/day; the web-app database is not its production store.

Evidence before action

Responsible use & architecture decisions

AGL202607071 · D = 1 · Small tier
Assigned feeder suffix 1 · Incident: six-hour South African regional outage during billing.

Table 3 · Responsible-use brief

Who could be harmed	Refuse or escalate	First position
Unmetered households receiving estimated bills	Refuse fabricated readings, unsupported cap values and automatic disconnection. Escalate missing evidence.	Require applicable approved cap and justification; preserve a human dispute route and correction record.
Low-income neighborhoods and rural feeders with late telemetry	Refuse to equate missing data or high-loss scores with theft. Human verification before field action.	Assess urban/rural and metered/estimated outcomes in Task 4; separate coverage effects from risk signals.
Customers whose routines can be inferred; people disputing disconnection	Refuse unrestricted identities and automatic enforcement. Escalate re-identification, foreign transfer and appeals.	Keep identity keys controlled in Abeokuta; minimize access; support explanation, challenge and human review.

ADR-01 · Managed PaaS for readings

Status: accepted proposal; not provisioned. **Context:** 19.2 million readings/day at the assigned production tier, bursty mobile delivery, nightly reconciliation and a three-engineer team.

Decision: managed ingestion/queueing, transformations and query services. Preserve the pseudonymized South African lake, subject to documented transfer safeguards. Put event-time outage alerting on an independent Nigerian path, subject to verified service availability.

Alternatives: IaaS shifts patching and platform operation to Sunbeam; self-managed Kubernetes/Kafka are not justified. FaaS alone does not resolve durable state and replay.

Consequences: reduce operating toil; retain provider dependence and foreign-exchange exposure. Sunbeam owns access, schema, quality, lawful processing, recovery tests and alert evidence. Use event-time windows, deduplication and dead-letter handling. Monitor end-to-end delay.

ADR-02 · Keep the identity key in Abeokuta

Status: accepted architectural refusal; vault is outside this demo. **Context:** re-identification enables household surveillance; pseudonymized consumption remains personal data.

Decision: retain the encrypted identity mapping and keys on Sunbeam-controlled infrastructure in Abeokuta, with restricted network access, separation of duties, approved case-based retrieval and tested encrypted backups. Do not move the mapping into the foreign lake.

Tradeoff: Sunbeam retains local host, patching, physical security, backup and recovery work. Assign ownership among existing engineers; location by itself does not prove compliance.

Deployment, continuity and evidence

Officer / reviewer	→ **Node API**	→ **SQLite volume**
Phone or browser	Docker / Dokku	Synthetic records only

During the six-hour regional outage, keep Nigerian operational alerts independent, queue exports, display stale status and pause evidence-dependent billing. Replay idempotently and reconcile on recovery. Do not disconnect automatically. These controls need failure testing in later tasks.

Revenue is NGN and cloud charges USD. A 30% fall in naira value means NGN/USD increases by $1 \div 0.7 = 1.4286$ under the stated interpretation. Reprice the Task 2 cost model; no provider quote is invented here.

Submission evidence: show actual running VirtualBox estate; the app deployed under Docker and Dokku; measured command counts; and service-model reasoning. Source includes Dockerfile, Procfile and full ADRs in decisions.md. This pack does not substitute for execution evidence.

References: supplied Task 1 brief; NERC Customer Protection Regulations 2023 and consumer complaints procedure (nerc.gov.ng); Nigeria Data Protection Act 2023 and NDPC transfer guidance (ndpc.gov.ng). Consult docs/DOCUMENTATION.md for direct links. Verify current applicable instruments before real customer processing.